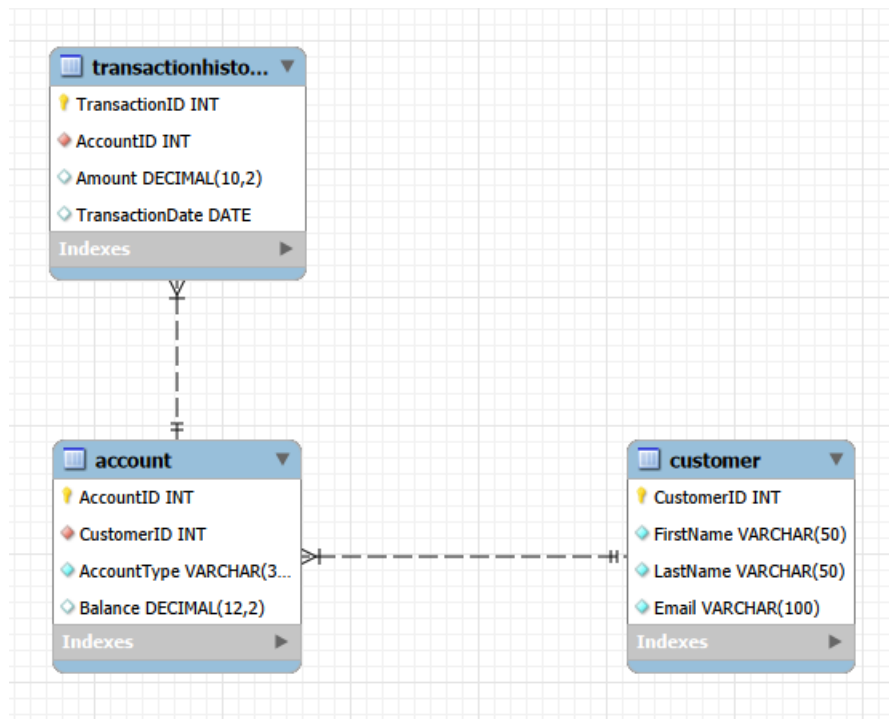




```
Query 1 x
Limit to 1000 r
32
33 -- 5. Check Schema Exists
34 • SELECT SCHEMA_NAME
35 FROM INFORMATION_SCHEMA.SCHEMATA
36 WHERE SCHEMA_NAME = 'BankDB';
37
```

Result Grid

SCHEMA_NAME
bankdb

Query 1 x

Limit to 1000

```
37
38 -- 6. Check Tables Exist
39 • SELECT TABLE_NAME
40 FROM INFORMATION_SCHEMA.TABLES
41 WHERE TABLE_SCHEMA = 'BankDB';
42
```

Result Grid | Filter Rows: | Export: | Wrap

TABLE_NAME
account
customer
transactionhistory

Query 1 x

Limit to 1000 r

```
43 -- 7. Check Column Structure of Customer
44 • SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY
45 FROM INFORMATION_SCHEMA.COLUMNS
46 WHERE TABLE_NAME = 'Customer'
47 AND TABLE_SCHEMA = 'BankDB';
48
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

COLUMN_NAME	DATA_TYPE	IS_NULLABLE	COLUMN_KEY
CustomerID	int	NO	PRI
Email	varchar	NO	UNI
FirstName	varchar	NO	
LastName	varchar	NO	

Query 1

Limit to 1000 r

```

48
49  -- 8. Check Primary Keys
50 • SELECT t.CONSTRAINT_NAME, k.COLUMN_NAME
51 FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS t
52 JOIN INFORMATION_SCHEMA.KEY_COLUMN_USAGE k
53 ON t.CONSTRAINT_NAME = k.CONSTRAINT_NAME
54 WHERE t.CONSTRAINT_TYPE = 'PRIMARY KEY'
55 AND t.TABLE_SCHEMA = 'BankDB';

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:

	CONSTRAINT_NAME	COLUMN_NAME
▶	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	index_name
	PRIMARY	stat_name
	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	index_name
	PRIMARY	stat_name
	PRIMARY	database_name
	PRIMARY	table_name
	PRIMARY	index_name
	PRIMARY	stat_name
	PRIMARY	OBJECT_INSTA...
	PRIMARY	OBJECT_INSTA...
	PRIMARY	OBJECT_INSTA...
	PRIMARY	LOGGED
	PRIMARY	LOGGED

Result 6

```
Query 1 x
Limit to 1000 r
56
57 -- 9. Check Foreign Keys
58 • SELECT TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME, REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME
59 FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE
60 WHERE TABLE_SCHEMA = 'BankDB' AND REFERENCED_TABLE_NAME IS NOT NULL;
61
```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME
account	CustomerID	account_ibfk_1	customer	CustomerID
transactionhistory	AccountID	transactionhistory_ibfk_1	account	AccountID

```
Query 1 x
Limit to 1000 r
61
62 -- 10. Check Indexes on Customer
63 • SHOW INDEX FROM Customer;
64
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
customer	0	PRIMARY	1	CustomerID	A	0				BTREE			YES	
customer	0	Email	1	Email	A	0				BTREE			YES	

```
65 -- 11. Insert Test Data into Customer
66 • INSERT INTO Customer (CustomerID, FirstName, LastName, Email)
67 VALUES
68 (1, 'Alice', 'Wong', 'alice.wong@bank.com'),
69 (2, 'Bob', 'Smith', 'bob.smith@bank.com');
70
71 -- 12. Insert Duplicate Email to Trigger Constraint Violation
72 -- This should fail
73 • INSERT INTO Customer (CustomerID, FirstName, LastName, Email)
74 VALUES (3, 'Charlie', 'Brown', 'alice.wong@bank.com');
75
```

#	Time	Action	Message
30	21:14:23	INSERT INTO Customer (CustomerID, FirstName, LastName, Email) VALUES (1, 'Alice', 'Wong', 'alice.wong@bank.com')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
31	21:15:07	INSERT INTO Customer (CustomerID, FirstName, LastName, Email) VALUES (3, 'Charlie', 'Brown', 'alice.wong@bank.com')	Error Code: 1062. Duplicate entry 'alice.wong@bank.com' for key 'customer.Email'

```
76 -- 13. Insert Data into Account
77 • INSERT INTO Account (AccountID, CustomerID, AccountType, Balance)
78 VALUES
79 (101, 1, 'Savings', 5000.00),
80 (102, 2, 'Checking', 1200.50);
81
82 -- 14. Insert Data into TransactionHistory
83 • INSERT INTO TransactionHistory (TransactionID, AccountID, Amount, TransactionDate)
84 VALUES
85 (1001, 101, 150.00, '2025-04-01'),
86 (1002, 101, -50.00, '2025-04-03');
87
88 -- 15. Test Foreign Key Violation (Invalid CustomerID)
89 -- This should fail
90 • INSERT INTO Account (AccountID, CustomerID, AccountType, Balance)
91 VALUES (103, 999, 'Fixed Deposit', 2000.00);
92
```

#	Time	Action	Message
32	21:15:53	INSERT INTO Account (AccountID, CustomerID, AccountType, Balance) VALUES (101, 1, 'Savings', 5000.00)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
33	21:15:59	INSERT INTO TransactionHistory (TransactionID, AccountID, Amount, TransactionDate) VALUES (1001, 101, 150.00, '2025-04-01')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
34	21:16:01	INSERT INTO Account (AccountID, CustomerID, AccountType, Balance) VALUES (103, 999, 'Fixed Deposit', 2000.00)	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('bankdb`.`account', CO...